

Data Driven Early Warning System for Aadhaar Enrolment and Demographic Update Anomalies

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Introduction

The Unique Identification Authority of India manages Aadhaar enrolment and update services at a national scale. Ensuring data quality, operational efficiency, and timely identification of regional irregularities is critical for effective governance. This project analyses anonymised Aadhaar enrolment and demographic update datasets to derive actionable insights that can support informed and proactive decision making.

Objectives of the Project

- Identify patterns and trends in Aadhaar enrolment and demographic updates
- Detect anomalies using data driven statistical methods
- Enable state and district level drill down for targeted operational review
- Propose a scalable early warning and monitoring framework

Problem Statement (Points)

- Aadhaar enrolment and demographic updates happen across many states and districts.
 - Raw numbers (counts) are difficult to compare because population size differs by region.
 - Some regions may show unusual update behaviour, but it is hard to detect early.
 - There is no simple way to identify which exact states or districts need attention.
 - Decision-makers need clear, data-based signals instead of manual checking.
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Proposed Solution (Points)

- Use anonymised and aggregated Aadhaar data to ensure privacy.
- Combine enrolment and demographic update data for analysis.
- Calculate an Update-to-Enrolment Ratio to compare regions fairly.
- Apply simple statistical methods to detect unusual or abnormal patterns.
- Drill down from state level to district level to locate the exact areas.
- Help authorities take timely, focused, and data-driven actions.

Datasets Used

- 1. Aadhaar Enrolment Dataset
- 2. Aadhaar Demographic Update Dataset

Both datasets are aggregated and anonymised. They include temporal and geographic attributes such as date, state, district, PIN code, and age wise counts, ensuring privacy while enabling meaningful analysis.

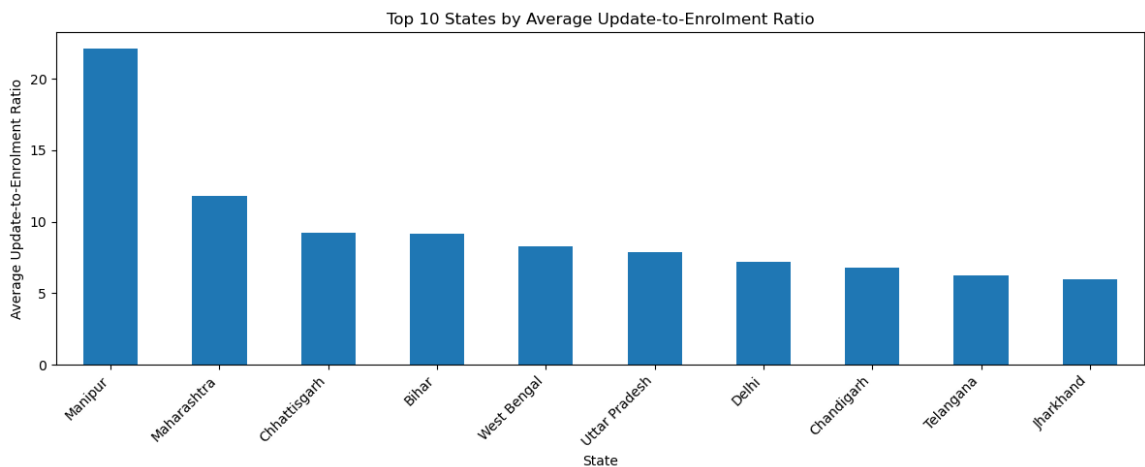
Methodology

The analysis follows a structured methodology:

- Consolidation of multiple dataset files
- Data cleaning and standardisation
- Aggregation to remove duplication
- Feature engineering using Update to Enrolment Ratio
- Statistical anomaly detection using Z score
- District level drill down for actionable insights

State-wise Comparison of Update-to-Enrolment Ratio

This analysis compares the average update to enrolment ratio across states to identify regional variations and highlight states requiring closer operational attention.



Anomaly Detection in Update-to-Enrolment Ratio

Z score based anomaly detection was applied to identify statistically significant deviations from the national average. Manipur was identified as a strong outlier, indicating disproportionately high demographic update activity.

	state	avg_update_enrolment_ratio	z_score
0	Manipur	22.148547	5.010489
1	Maharashtra	11.792391	2.073258
2	Chhattisgarh	9.198247	1.337502
3	Bihar	9.144124	1.322151
4	West Bengal	8.285426	1.078606
5	Uttar Pradesh	7.898373	0.968830
6	Delhi	7.166482	0.761249
7	Chandigarh	6.790667	0.654660
8	Telangana	6.212322	0.490629
9	Jharkhand	5.964104	0.420229

District-Level Analysis for Anomalous State

Following the identification of Manipur as an anomalous state, a district level drill down was conducted to localise the source of the elevated ratio.

High contributing districts in Manipur:

- Bishnupur – 49.26
- Thoubal – 47.21
- Imphal East – 30.83
- Imphal West – 29.21
- Kakching – 13.92
- Ukhrul – 12.80
- Chandel – 10.11
- Senapati – 8.79
- Churachandpur – 7.28
- Tamenglong – 7.04

Key Insights

- High update ratios are concentrated in specific districts rather than across the entire state
- Patterns suggest localised operational or process-related factors
- Ratio-based metrics enable fair comparison across regions with different population scales

Recommendations

- Conduct targeted operational reviews in high ratio districts
- Improve enrolment time data validation and operator training
- Use update to enrolment ratio as an early warning indicator
- Deploy district level monitoring dashboards for continuous oversight

Conclusion

This project demonstrates how anonymised Aadhaar data can be transformed into actionable intelligence. The proposed early warning framework supports proactive governance, efficient resource allocation, and continuous monitoring while maintaining data privacy.